

Thursday, July 2, 2026 1:35 PM

random sample : n observations

$$\text{Sample mean : } \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

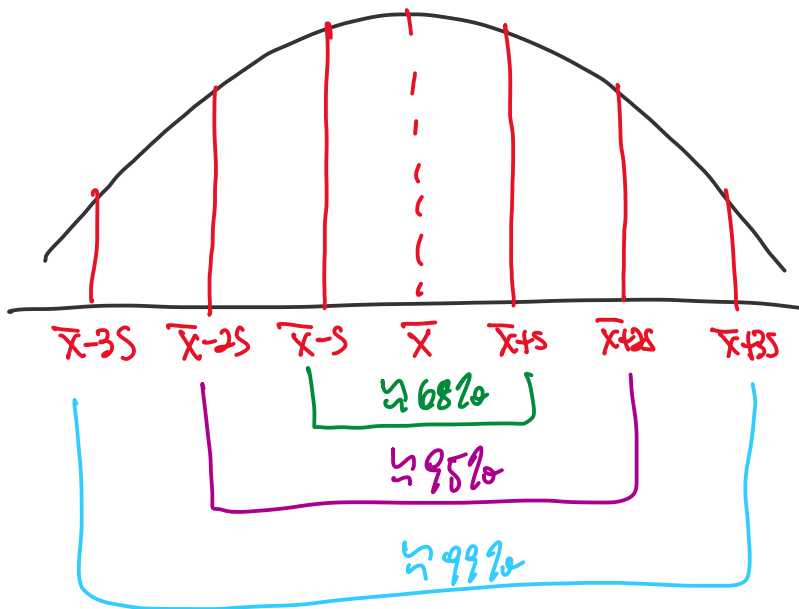
unbiased estimator of μ

$$\text{sample variance : } s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

 $n-1$ degrees of freedomunbiased estimator of σ^2

$$\text{Sample standard deviation : } s = \sqrt{s^2}$$

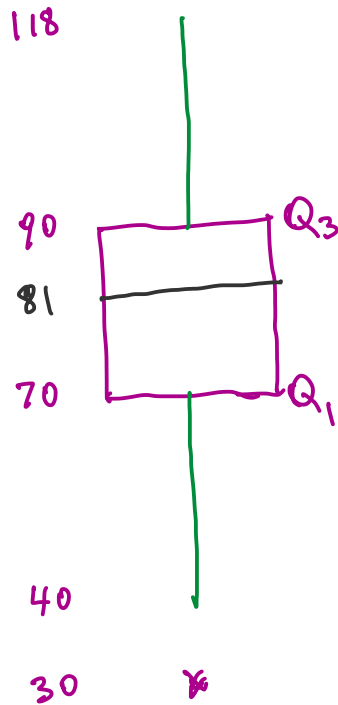
Empirical Rule



$$Z = \frac{x - \bar{x}}{s}$$

 $|Z| \geq 3$: outlier $2 \leq |Z| < 3$: somewhat an outlier

Boxplot
 $Q_3 = Q_U$: 3rd Quartile: 75th percentile
 $Q_1 = Q_L$: 1st Quartile: 25th percentile
 $Q_3 - Q_1$: IQR



Simple Events

- Simple events are mutually exclusive events that represent the most basic outcomes of an experiment.
- The sample space of an experiment is the collection of all of its simple events.
- For example, if two fair coins are tossed then what are the simple events and thus what is the sample space?

$\{HH, HT, TH, TT\}$

0 Tails $\longleftrightarrow$ HH
 1 Tail $\begin{cases} HT \\ TH \end{cases}$
 2 Tails $\longleftrightarrow$ TT

Intersections and Unions

- Intersection**

- A and B occur simultaneously
- When both events A and B occur, we refer to this as the intersection of A and B denoted by $A \cap B$.

- Union**

- Either A or B (including the case where both A and B occur simultaneously)
- When either A or B or both events occur we refer to this as the union of A and B denoted by $A \cup B$

A: Even #

$A = \{2, 4, 6\}$

B: Greater than 4

$B = \{5, 6\}$

$A \cap B = \{6\}$

$$A \cup B = \{2, 4, 5, 6\}$$

Conditional Probability

- What is the difference between the following two problems?

1. What is the probability that the sum of two dice is 5? $\frac{1}{9}$

2. Given that the first die is less than 3, what is the probability that the sum of the two dice is equal to 5? $\frac{1}{6}$

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Example with Sample Space

- You roll two fair dice.
- Let A be the event the sum of two dice is 5.
- Let B be the event the first die is less than 3.
- Let C be the event the sum of two dice is 7.

$$1. P(A) = \frac{1}{9}$$

$$2. P(B) = \frac{1}{3}$$

$$3. P(C) = \frac{1}{6}$$

$$4. P(A|B) = \frac{1}{6}$$

$$8. P(C|B) = \frac{1}{6}$$

$$5. P(A \cap B) = P(A|B)P(B) = \frac{1}{18}$$

$$9. P(C \cap B) = P(C|B)P(B) = \frac{1}{18}$$

$$6. P(A \cup B) = P(A) + P(B) - P(A \cap B) =$$

$$10. P(C \cup B) = P(C) + P(B) - P(C \cap B) =$$

$$7. P(B|A) \left[\frac{1}{9} + \frac{1}{3} - \frac{1}{18} = \frac{7}{18} \right]$$

$$11. P(B|C) = \frac{1}{3}$$

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$$P(B|A)P(A) = P(A \cap B)$$

$$\frac{1}{18} / \frac{1}{9} = \frac{1}{2}$$

